

Nonconvexity of fixed odd-power sum-of-squares cones

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Abstract

For a fixed odd integer q , consider the nonnegative real homogeneous forms whose q th powers are sums of squares. We give a negative answer to the general-dimensional convexity question recorded by Reznick in 2023. For sextics and $q = 3$, the cone is nonconvex in the explicit finite dimension $3 \cdot 10^{62}$. For every odd $q \geq 3$, it is nonconvex in some finite dimension depending on q . The proof combines a classical modified Motzkin form with an SOS cube, separation from the SOS cone, positivity of product truncated functionals, and elementary distinct-index counting. An explicit rational moment functional gives the displayed dimension; its 220×220 local positivity certificate can be checked without floating-point arithmetic. No ternary or smallest-dimension conclusion is made.

1 The question and the result

Let $\mathbb{R}[x_1, \dots, x_n]_m$ be the real forms of degree m , where m is positive and even. For a positive odd integer q , set

$$\mathcal{C}(n, m, q) = \{f \in \mathbb{R}[x_1, \dots, x_n]_m : f \geq 0 \text{ on } \mathbb{R}^n, \quad f^q \in \text{SOS}\}.$$

Here SOS means a finite sum of squares of real polynomials. Nonnegativity is automatic from the second condition when q is odd. These sets are closed and stable under multiplication by nonnegative scalars; whether they are convex is a separate question. Reznick asks this for general n, m, q in his 2023 Oberwolfach contribution [1, p.779]. The same question appears in [2, §6].

Theorem 1. *Put $N = 10^{62}$ and $n = 3N$. On disjoint variable triples (x_i, y_i, z_i) , define*

$$p_i = x_i^4 y_i^2 + x_i^2 y_i^4 + z_i^6 - x_i^2 y_i^2 z_i^2 \quad (1 \leq i \leq N).$$

Each p_i belongs to $\mathcal{C}(n, 6, 3)$, whereas $N^{-1} \sum_{i=1}^N p_i$ does not. Thus $\mathcal{C}(3 \cdot 10^{62}, 6, 3)$ is not convex. Moreover, for every fixed odd $q \geq 3$ there is a finite $n(q)$ for which $\mathcal{C}(n(q), 6, q)$ is not convex.

The explicit dimension is a sufficient bound, with no optimality claim. The theorem refutes the general convexity assertion; it does not settle the ternary fixed-power question, classify parameter triples, or give a common dimension for all odd exponents. Each counterexample persists on adding unused variables, because restriction of an SOS to those variables equal to zero is again SOS. The case $q = 1$ and classical cases in which every nonnegative form is SOS remain consistent with the theorem.

The ingredients are established. The exact modified Motzkin seed is due to Berg–Christensen–Jensen [3, Lemma 1]; its SOS cube is recorded in [1, 2]. Closure and positive-functional separation

are classical as well [3, Theorems 3–4]. Product pseudoexpectation positivity is explicit in Barak–Kelner–Steurer [4, Lemma A.5]. Distinct-index versus collision asymptotics appear, for example, in [5, Lemma 2.1] and [6, Proposition 6.4]. We apply these tools to obstruction at one *prescribed* odd power and supply an explicit finite certificate. The convexity of the union over odd powers in [2, Theorem 5.3] and [7, Theorem 2.2] allows the exponent to increase and does not imply fixed-power convexity.

2 Disjoint-copy obstruction at a fixed odd power

We write $\mathbb{R}[u]_{\leq r}$ for polynomials of total degree at most r in a fixed finite variable block u .

Lemma 1. *Let $p \in \mathbb{R}[u]_m$, let q be positive and odd, and put $d = qm/2$. Suppose $L : \mathbb{R}[u]_{\leq 2d} \rightarrow \mathbb{R}$ satisfies*

$$L(1) = 1, \quad L(h^2) \geq 0 \text{ (deg } h \leq d), \quad a_1 := L(p) < 0.$$

For all sufficiently large finite integers s , the polynomial $(p(u_1) + \cdots + p(u_s))^q$ on disjoint copies of u is not SOS.

Proof. The local moment matrix $M = (L(u^{\alpha+\beta}))_{|\alpha|, |\beta| \leq d}$ is positive semidefinite. On polynomials of degree at most $2d$ in *each* of the s blocks, define a product functional on monomials by

$$L_s \left(\prod_{i=1}^s u_i^{\alpha_i} \right) = \prod_{i=1}^s L(u^{\alpha_i}).$$

The matrix controlling $L_s(H^2)$ for all H of *total* degree at most d is a principal submatrix of $M^{\otimes s}$, hence is positive semidefinite. Consequently L_s is nonnegative on every sum of such squares. This is the standard product-positivity construction [4]. All spaces and products here are finite; no representing measure is assumed.

Put $a_j = L(p^j)$ for $0 \leq j \leq q$, so $a_0 = 1$, and let $S_s = \sum_{i=1}^s p(u_i)$. Expansion by partitions of the q labelled factors gives

$$L_s(S_s^q) = \sum_{\pi \in \text{Part}([q])} (s)_{|\pi|} \prod_{B \in \pi} a_{|B|}, \quad (s)_r = s(s-1) \cdots (s-r+1). \quad (1)$$

For each partition, assigning its distinct blocks to distinct copies has $(s)_{|\pi|}$ choices. The unique partition with q blocks is the singleton partition. Thus the expression is a polynomial in s of degree q , with leading coefficient $a_1^q < 0$; all collisions contribute lower degree. It is strictly negative for all sufficiently large integers s .

An SOS polynomial of degree at most $2d$ can involve only square summands of degree at most d : otherwise its highest-degree homogeneous part would be a nonzero sum of squares, and could not cancel. Therefore the negative value excludes every SOS representation of S_s^q . $\square$

Lemma 2. *If a form p is not SOS and q is positive and odd, then with $d = qm/2$ there exists an L satisfying the hypotheses of Lemma 1.*

Proof. The cone of SOS polynomials of degree at most $2d$ is closed. Indeed, let v be the vector of all monomials of degree at most d and write a coefficient-convergent SOS sequence as $v^T Q_j v$, $Q_j \succeq 0$. For a standard Gaussian integral Γ , $G = \Gamma(vv^T)$ is positive definite. Coefficient convergence bounds $\Gamma(v^T Q_j v) = \text{tr}(Q_j G)$, which bounds $\text{tr} Q_j$, and hence the matrices Q_j . A convergent subsequence yields a positive semidefinite Gram matrix for the limit.

The same highest-degree argument used above shows that a non-SOS form p is outside this larger finite-degree SOS cone. Finite-dimensional separation from a closed convex cone gives a functional L_0 nonnegative on that cone with $L_0(p) < 0$. For sufficiently small $\varepsilon > 0$, $L_0 + \varepsilon\Gamma$ still has negative value on p and has positive definite moment matrix. Its value on 1 is positive. Dividing by that value gives the required L . This is classical SOS duality [3]. $\square$

The power map $f \mapsto f^q$ is continuous in the coefficient topology. The closedness just proved therefore also gives closedness of each fixed-power set $\mathcal{C}(n, m, q)$ as the inverse image of the fixed-degree SOS cone.

In particular, if a non-SOS nonnegative form p has an SOS q th power, its disjoint copies belong to the same ambient $\mathcal{C}$, whereas a sufficiently large finite average does not. Convexity implies closure under every finite convex combination by induction, so that average refutes convexity. If a violating pair is desired, take the least failing prefix of the finite sum in the same ambient variable space; its preceding prefix and next summand are inside the cone. This asserts existence of a pair, without assigning an SOS power to any untested prescribed prefix.

3 The classical seed

Set

$$p = x^4y^2 + x^2y^4 + z^6 - x^2y^2z^2. \quad (2)$$

AM–GM gives $x^4y^2 + x^2y^4 + z^6 \geq 3x^2y^2z^2$, hence $p \geq 0$. The dehomogenization $p(x, y, 1)$ is exactly the example in [3, Lemma 1]. Below we also independently certify $p \notin \text{SOS}$ by a positive-on-squares functional with $L(p) = -1$. The credited SOS identity for p^3 [1, 2] is displayed in Appendix A and checked by exact polynomial expansion in the supplement. For every odd $q \geq 3$,

$$p^q = p^3(p^{(q-3)/2})^2 \in \text{SOS}.$$

Lemmas 1 and 2 therefore prove the second assertion of Theorem 1. The functional may depend on q ; no degree 18 functional is reused outside its domain.

4 An explicit rational functional for the cube

For $\alpha \in \mathbb{N}^3$, let g_α be the standard Gaussian moment: it is zero if any coordinate is odd, and otherwise $g_\alpha = \prod_{j=1}^3 (\alpha_j - 1)!!$, with $(-1)!! = 1$. We define $\mu_\alpha = L(x^{\alpha_1}y^{\alpha_2}z^{\alpha_3})$ for $|\alpha| \leq 18$. Set all moments with an odd coordinate to zero. For even coordinates, put $\mu_0 = 1$, and $\mu_\alpha = g_\alpha/1024$ when $|\alpha| = 2$ or 4. For degree 6, write $\alpha = 2(a, b, c)$ and use

(a, b, c)	$(3, 0, 0)$	$(0, 3, 0)$	$(2, 1, 0)$	$(1, 2, 0)$	$(0, 0, 3)$
$\mu_{2(a,b,c)}$	2^{24}	2^{24}	1	1	1
(a, b, c)	$(1, 1, 1)$	$(1, 0, 2)$	$(0, 1, 2)$	$(2, 0, 1)$	$(0, 2, 1)$
$\mu_{2(a,b,c)}$	4	32	32	4096	4096

For $|\alpha| = 2r$, $4 \leq r \leq 9$, put $\mu_\alpha = T_r g_\alpha$, where

$$\frac{r}{T_r} \mid \begin{array}{cccccc} 4 & 5 & 6 & 7 & 8 & 9 \\ 10^{16} & 10^{34} & 10^{53} & 10^{73} & 10^{94} & 10^{116} \end{array}. \quad (3)$$

This table specifies the entire order-nine moment matrix, with $\binom{12}{3} = 220$ monomials. Coordinate parity splits it into eight blocks of size at most 35.

Proposition 1. *The functional specified above is positive on every nonzero square of degree at most 18: $L(h^2) > 0$ for $0 \neq h$, $\deg h \leq 9$.*

Proof. Here is a compressed exact rational certificate and its verification rule. Order multi-indices lexicographically by increasing first coordinate, then second, then third, within the set $|\alpha| \leq r$; retain this order in each parity block. At order 3 the leading principal minors are as follows. The entries in each row list all such minors for that block.

Parity	Leading principal minors at order 3
000	1, 3071/1048576, 2047/268435456, 5117/274877906944
001	1/1024, 1015/1048576, 3109055/1048576, 1056109395/131072
010	1/1024, 32767/1048576, 16642474743/32768, 257328824345/1048576
100	1/1024, 32767/1048576, 16359/1048576, 257328824345/1048576
011,101,110	1/1024 each
111	4

They are positive, so Sylvester's criterion establishes order 3 positivity. For $r = 4, \dots, 9$, split each block with new degree- r monomials into old and new rows. It has the form

$$\begin{pmatrix} A & B \\ B^T & T_r G \end{pmatrix}, \quad G = (g_{\alpha+\beta})_{\substack{|\alpha|=|\beta|=r \\ \alpha, \beta \text{ of the given parity}}}.$$

The Gaussian matrix G is positive definite. The lower-degree block A is positive definite by induction; B uses already specified moments. Let $C = B^T A^{-1} B$. Exact rational calculations give

$$0 \leq \text{tr}(G^{-1}C) < T_r \quad (4)$$

for each of the four extending blocks at each order. The supplement's rational witness file `TENSOR_MOMENT_CERTIFICATE.json` lists all 24 exact fractions, the old/new dimensions and strict bounds. Its standalone checker rebuilds A, B, G from the displayed moments, solves the rational systems and checks these fractions and inequalities; supplied positivity flags are not used. It also independently computes all 220 positive pivots of the full local blocks and verifies the exact LDL reconstruction.

Since $C \succeq 0$, the largest eigenvalue of $G^{-1/2}CG^{-1/2}$ is at most its trace. Equation (4) therefore implies $T_r G - C \succ 0$. The Schur complement makes every enlarged block positive definite. Blocks with no new rows are unchanged. Induction proves the proposition. The finite computations in this step are exact rational certificate checks, with no numerical eigensolver or SDP. $\square$

5 The finite-dimensional counterexample

Exact expansion using the moment table yields

$$a_1 = L(p) = -1, \quad a_2 = L(p^2) = 11292 \cdot 10^{53}, \quad a_3 = L(p^3) = 35039520 \cdot 10^{116}. \quad (5)$$

The first value is $1 + 1 + 1 - 4$. For the other two, homogeneity means all moments have degree 12 or 18, respectively, so these are Gaussian polynomial moments times T_6 or T_9 . The checker expands both polynomials exactly. Proposition 1 and $L(p) = -1$ also prove that p is not SOS.

For $S_N = \sum_{i=1}^N p_i$, the cubic case of (1) is

$$L_N(S_N^3) = Na_3 + 3N(N-1)a_2a_1 + N(N-1)(N-2)a_1^3.$$

Since $a_1 = -1$ and $a_2 \geq 1$, we obtain

$$L_N(S_N^3) \leq N\{a_3 - (N^2 - 1)\} < 0,$$

where the strict inequality follows from

$$35039520 \cdot 10^{116} < 10^{124} - 1 = N^2 - 1 \quad (N = 10^{62}).$$

Lemma 1's product positivity excludes every SOS for S_N^3 , and positive scaling excludes one for $(S_N/N)^3$. Each p_i^3 has the credited SOS identity. Thus the average is a finite convex combination of elements of $\mathcal{C}(3N, 6, 3)$ outside that set, proving the first assertion of Theorem 1.

The functional L is not a measure on real points: its negative value on the nonnegative p prevents that interpretation. Only positivity on the required squares is asserted. The argument neither constructs an enormous matrix numerically nor assumes an infinite product probability measure.

6 Verification, priority and disclosure

The accompanying verification archive contains this standalone source, the complete rational certificate, a Python-standard-library checker, expected deterministic output, hash/provenance records, the claim scope and a dated priority note. The checker uses integers and rational fractions to verify the local moment matrices, all initial minors and Schur traces, the credited cube identity, the three seed moments and the exact finite cubic sign. These computations support the finite certificate; the arbitrary finite product argument and the every-odd- q assertion are proved analytically above. No formal proof-assistant certificate is claimed.

Two independent extensive bounded priority audits identified no exact earlier fixed-power result in the sources examined. Their source versions, reading extents and remaining gaps are disclosed in the supporting note. The full final Papp–Alizadeh 2013 article and Schmüdgen 1979 body were unavailable; a captured Choi–Lam–Reznick 1995 scan was unread and unused. Classical tensor/separation ingredients have independently accessible primary proofs. Bibliographies, citation indexes and talk coverage have limits. This is not a worldwide-firstness guarantee or a certificate of present openness. The claim concerns the general question recorded in 2023.

AI tools were used extensively in derivation, literature comparison, reproduction, writing and adversarial verification. This is an unrefereed preprint and has not received human peer review. Mathematical and source claims remain open to independent correction. The author metadata does not imply affiliation with a research institution.

A The credited cube identity

For the seed (2), put

$$\begin{aligned} h_1 &= x^5 y^4 - x^3 y^4 z^2, & h_2 &= x^4 y^5 - x^4 y^3 z^2, \\ h_3 &= x^4 y^2 z^3 - x^2 y^2 z^5, & h_4 &= x^2 y^4 z^3 - x^2 y^2 z^5, \\ h_5 &= x y^2 z^6 - x^3 y^4 z^2, & h_6 &= x^2 y z^6 - x^4 y^3 z^2, \\ h_7 &= x^2 y^4 z^3 - x^4 y^2 z^3, & h_8 &= x^4 y^5 - x^2 y z^6, \\ h_9 &= x^5 y^4 - x y^2 z^6. \end{aligned}$$

Then the identity in [2], specialized to this seed, is

$$\begin{aligned} p^3 = & \frac{3}{2} \sum_{j=1}^9 h_j^2 + (z^9 - 2x^2y^2z^5)^2 + (xyz^7 - 2x^3y^3z^3)^2 \\ & + (x^3y^6 - 2x^3y^4z^2)^2 + (x^3y^5z - 2x^3y^3z^3)^2 \\ & + (x^6y^3 - 2x^4y^3z^2)^2 + (x^5y^3z - 2x^3y^3z^3)^2 + 2(x^3y^3z^3)^2. \end{aligned}$$

Every square factor has degree 9 and every weight is positive. This is the modified coefficient-1 seed, not the coefficient-3 original Motzkin form.

References

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